PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

FORM FOUR NORTH WESTERN ZONE MOCK EXAMINATION SEP. 2024

032/2A CHEMISTRY 2B (ACTUAL PRACTICAL)

TIME; 2:00 HRS **01/10/2024 TUESDAY**

Instructions

1. This questions consists of two questions
2. Answer all questions
3. Question 1 carries 25 marks and question 2 carries 25 marks
4. Write your registration number on every page of your answer sheets
5. You may use the following constants

   H= 1, Cl= 35.5

   1 litre = $1dm^3$= $1000cm^3$

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1. You are provided with the following solutions
   A. Containing $4.6g/dm^3$ of HX
   B. Containing 1.00g of sodium hydroxide in $0.25dm^3$ of solutions. Methyl orange indicator may used

Questions

(a) With state symbols, write a balanced chemical equation for the reaction between A and B

(b) Titrate solution A (in burette) against solution B (in conical flask) using two drops of your indicator and obtain three titre values.

(c) (i) ……cm^3 of A and required ……cm^3 of B for complete reaction

(ii) Showing your procedure clearly, identify element X in HX

2. Salt sample Q is a simple salt. Carryout the experiments described below. Record your observations and make appropriate inferences and hence identify the cation and anion present in sample.

S/N	EXPERIMENT	OBSERVATION	INFERENCE
a)	Observe the appearance of solid sample Q		
b)	Heat a little solid sample Q in a dry test tube		
c)	Perform flame test on solid sample Q		
d)	Add a few drops ofconc.H_2SO_4 acid into a solid sample Q in a test tube and warm the contents gently		
e)	Dissolve a little sample Q in water and divide the solution in four portions		
	i/ To the first portion add Na0H solution till in excess		
	ii/ To the second portion add ammonia solution till in excess		
	iii/ To the third portion add freshly prepared $FeS0_4$ solution followed by careful addition of concentrated H_2S0_4 along the side of test tube		
	iv/ To the fourth portion add potassium hexcyanoferate (II)		

Conclusion;

i) The cation in sample Q is ----------

ii) The anion in sample Q is ----------

iii) The formula of salt Q is ----------

iv) The name of the salt obtained in (iii) is……

v) A balanced chemical equation taking place in experiment **b** is ……

vi) The type of chemical reaction formed in (v) above is…….